Liptis Nutrition NeoPed™ — Dual-Engine Clinical Platform

A comprehensive clinical decision-support system for neonatologists and pediatricians in NICUs and outpatient clinics. Unifies precision enteral feeding calculations with Fenton 2013 and WHO 2006 age-corrected growth trajectory monitoring to eliminate math errors and ensure physiological catch-up growth.

THE BEDSIDE CLINICAL PROBLEM VS. THE NEOPED™ DUAL SOLUTION

⚠ Critical Bedside Challenges in Preterm Care

- **Complex Preterm Age Transformations:** Calculating exact Chronological Age, Post-Menstrual Age (PMA), and Corrected Age (CCA) for premature infants is prone to calendar errors and leap-year miscalculations.
- **High Risk of Dilution & Solute Errors:** Converting target fluids into exact scoops, grams of powder, and water dilution takes 10–15 minutes per baby with high risk of hyperosmolar stress or necrotizing enterocolitis (NEC).
- **Delayed Growth Failure Detection:** Without automated LMS Z-score tracking and weight velocity calculation, extrauterine growth restriction (EUGR) is often diagnosed too late.

✓ How NeoPed™ Solves Both in Under 10 Seconds

- **Engine 1 (Tab 1) — Fenton/WHO Growth Plotter:** Automatically derives GA, CA, PMA, and Corrected Age; plots Weight, Length, and Head Circumference on Fenton 2013 curves using Cole & Green LMS Z-scores, switching seamlessly to WHO 2006 at term.
- **Engine 2 (Tab 2) — Feeds & Macro Engine:** Doctor enters weight and fluid allowance; instantly derives total 24h volume, exact scoops of Pediamil® LBW, sterile water volume, and bedside schedules (q3h, q2h, continuous pump).
- **Live ESPGHAN 2022 Accordance:** Real-time audit of energy, 60:40 protein, P:E ratio, lipids, and bone minerals.

THE UNIFIED CLINICAL WORKFLOW FOR THE ATTENDING PHYSICIAN

1 Tab 1: Demographics, Age & Anthropometry

The physician enters Sex, Gestational Age (e.g. 28w + 2d), DOB, and DOM. The engine computes exact **Chronological Age**, **Post-Menstrual Age (PMA)**, and **Corrected Age**, while evaluating **Weight, Length, and Head Circumference** percentiles and LMS Z-scores.

2 Tab 2: Clinician Prescription & Scoops Engine

Doctor specifies prescribed fluid allowance (135–200 mL/kg/d; typical 150–160 mL/kg/d). The engine determines total 24h volume (mL), exact daily scoops of Pediamil® LBW, and required sterile water volume at standard 15.0% w/v dilution.

3 Live ESPGHAN 2022 Accretion Audit

Delivered intake is cross-checked in real time against ESPGHAN 2022 guidelines for Energy (115–140 kcal/kg/d), Protein (3.5–4.0 g/kg/d), P:E ratio (2.8–3.6 g/100 kcal), Lipids (4.8–8.1 g/kg/d), Calcium (120–140 mg/kg/d), Phosphorus, and Iron.

4 Bedside Orders & Printable Hospital Sheet

Outputs exact bedside aliquots for 8 feeds (q3h), 12 feeds (q2h), or continuous syringe pump infusion, with 1-click export of an official **Hospital Feeding Order Sheet (PDF)** ready for attending physician signature and medical records.

KEY PILLARS FOR KEY OPINION LEADERS & DEPARTMENT HEADS



Zero Clinical Error

Pure deterministic mathematics verified by 243 automated tests. Prevents calculation drift and solute overload.



Ward & Mobile Optimized

Responsive interface engineered for smartphones during rounds, tablets, and hospital desktop workstations.



100% Client-Side Privacy

Zero patient identifiers or records leave the device. Full compliance with HIPAA and EU GDPR standards.

THE 4 PRETERM AGE TRANSFORMATIONS EXPLAINED FOR DOCTORS

Accurate age transformation is the foundation of neonatal care. Treating a 28-week preterm infant by postnatal age alone leads to severe misinterpretation of growth, renal clearance, and neurodevelopment. NeoPed™ automatically calculates all 4 clinical age metrics:

Gestational Age (GA)

28w + 2d

At Birth

Completed weeks & days from mother's LMP. Defines prematurity tier: ELBW (<1000g), VLBW (<1500g), or LBW (<2500g).

Chronological Age (CA)

14 days

2.0 weeks (0.5 mo)

Elapsed postnatal time since delivery (DOM − DOB). Governs early electrolyte shifts and transitional feeding phases.

Post-Menstrual Age (PMA)

30.3 weeks

GA + Postnatal Age

Total maturational age. Governs Fenton preterm curve routing (22–50w), suck-swallow coordination, and apnea of prematurity.

Corrected Age (CCA)

-9.7 weeks

CA − (40w − GA)

Age adjusted for prematurity. Crucial for WHO 2006 plotting, developmental milestones, and avoiding false diagnosis of microcephaly.

THE 3 CORE ANTHROPOMETRIC PARAMETERS & COLE & GREEN LMS METHOD

1. Body Weight (g / kg)

- Primary somatic growth indicator.
- Evaluated via LMS: calculates exact continuous Z-score (e.g. −1.28 SD) and centile (e.g. 10th percentile).
- Alerts if weight drops <10th percentile (EUGR / Extrauterine Growth Restriction).

2. Crown-Heel Length (cm)

- Skeletal linear growth marker.
- Evaluates lean tissue accretion and protein sufficiency.
- Essential for calculating **Weight-for-Length proportionality** and distinguishing symmetrical vs asymmetrical IUGR.

3. Head Circumference (cm)

- Direct proxy for brain growth and neurodevelopment.
- Disproportionate head-sparing or microcephaly is flagged immediately.
- Tested against sex-specific Fenton 2013 and WHO 2006 reference tables.

PATEL ET AL. (2005) 2-POINT EXPONENTIAL WEIGHT VELOCITY (G/KG/DAY)

The Gold Standard Preterm Growth Velocity Model:

TARGET: 15–20 G/KG/DAY

Traditional linear velocity ($Delta W / Delta t$) artificially penalizes larger infants. NeoPed™ implements the peer-reviewed **Patel et al. (2005) exponential 2-point model**: $GV = [1000 \times \ln(W_2 / W_1)] / (D_2 - D_1)$. This tracks physiological daily weight accretion relative to current mass, ensuring true catch-up growth mirroring 3rd-trimester fetal rates.

DUAL GROWTH DATASET ROUTING & SEAMLESS TRANSITION

MATURATIONAL HORIZON	REFERENCE STANDARD	PRIMARY POPULATION	CLINICAL APPLICATION IN NEOPED™
22.0 to 50.0 weeks PMA	**Fenton 2013 Preterm Curves** (PMID: 23601190)	Preterm infants born <37 weeks GA	Intrauterine growth trajectories for Weight, Length, HC with sex-specific Cole & Green LMS Z-scores.
> 50.0 weeks PMA (0–24m CCA)	**WHO 2006 Child Growth Standards** (PMID: 16817681)	Infants at term-equivalent age	Seamless post-term transition using Corrected Age (CCA) to evaluate optimal infant neurodevelopmental growth.
Serial Longitudinal Tracking	**Multi-Visit Trajectory Engine**	NICU stay & outpatient follow-up	Tracks $Delta Z$-scores across visits to alert clinicians if weight or HC deviates by >1.0 standard deviation.

Key Opinion Leader Takeaway: True Multi-Metric Evaluation

NeoPed™ provides doctors with a comprehensive anthropometric assessment (Weight, Length, Head Circumference, and Growth Velocity) alongside precise age derivations (GA, CA, PMA, CCA). This enables neonatologists to identify nutritional failure weeks before standard paper charts.

THE DOCTOR'S PRESCRIPTION TO BEDSIDE RECONSTITUTION FLOW

NeoPed™ bridges the gap between the doctor's fluid prescription and the nursing staff's bedside bottle preparation. Doctors no longer need to calculate scoops or powder grams manually.

1. Prescribed 24h Volume

202.5 mL / day

Derived from patient weight (1,350g) and doctor's prescribed fluid allowance (150 mL/kg/d). *1.35 kg × 150 mL/kg/d = 202.5 mL.*

2. Daily Pediamil® LBW Scoops

6.1 scoops / day

Standard 15.0% w/v dilution: 30.4g powder / 5.0g per level scoop = **6.1 scoops/day** in **182.3 mL sterile water**.

3. Standard Water Dilution

182.3 mL water

Calibrated reconstitution: 1 scoop per 30 mL sterile water (3 level scoops + 90 mL sterile water = 100 mL prepared feed).

STANDARD FORMULA RECONSTITUTION GUIDE (WHO/FAO SAFE PROTOCOL)

RECONSTITUTION PARAMETER	PEDIAMIL® LBW STANDARD DILUTION	CLINICAL PREPARATION RULE	SAFETY & MICROBIOLOGICAL STANDARD
Concentration (% w/v)	15.0 g / 100 mL feed	3 level scoops per 90 mL sterile water yields 100 mL feed	Calibrated osmolality safe for preterm immature intestinal mucosa.
Manufacturer Scoop Size	5.0 g per level scoop	Use only enclosed LBW scoop; level with flat edge, do not pack	Prevents hyperconcentration and hyperosmolar renal load.
Sterile Water Ratio	30 mL water per scoop	90 mL water per 3 scoops (180 mL water for 6 scoops)	Water volume = Daily Volume × 0.9 (replaces water displacement).
Reconstitution Temp.	> 70°C sterile water	Boiled water cooled to >70°C before adding powder	Inactivates *Cronobacter sakazakii* per WHO/FAO PIF guidelines.

BEDSIDE FEEDING TIMETABLE: PRACTICAL 1,350G VLBW CASE EXAMPLE

Patient Demographics: 1,350g VLBW Preterm Infant • Fluid Allowance: **150 mL/kg/day** • Total 24h Volume: **202.5 mL/day** (~**6.1 scoops/day** in **182.3 mL sterile water**).

FEEDING ROUTE / SCHEDULE	FREQUENCY / 24H	VOLUME PER FEED	SCOOPS OF LBW	NURSING BEDSIDE ADMINISTRATION GUIDELINE
q3h Enteral Bolus Standard NICU convalescent feed	8 feeds / day	25.3 mL / feed	~0.8 scoop / feed	Administer by gravity bolus over 15–20 minutes. Check gastric residual volume prior to feed if clinically indicated.
q2h Enteral Bolus For smaller gastric volume capacity	12 feeds / day	16.9 mL / feed	~0.5 scoop / feed	Smaller aliquots reduce gastric distension, postprandial splanchnic steal, and reflux in high-risk preterm infants.
Continuous Enteral Infusion Severe intolerance or ELBW infant	Continuous 24h	8.4 mL / hour	–	Infuse continuously via syringe pump over 24 hours. Invert syringe periodically to prevent lipid phase separation.

TERM STEP-DOWN MILESTONE: TRANSITION TO PEDIAMIL® 1

Graduation Safeguard at 3,500g (Term-Equivalent Mass)

Preterm formulas have high protein and mineral densities necessary for rapid intrauterine catch-up growth. However, once an infant reaches **3,500g**, continuing high-density preterm formula risks excessive renal solute burden and adiposity. NeoPed™ automatically guides the doctor to transition to **Pediamil® 1** (68.5 kcal, 1.49g protein/100 mL, 13.7% w/v dilution, 4.57g scoop) to ensure smooth transition to standard term nutrition.

JPGN 2023;76(2):248–268 • PMID: 36705703

WHAT DOES THE PRESCRIBED VOLUME & SCOOPS DELIVER TO THE INFANT?

When an attending physician prescribes **Pediamil® LBW** at standard physiological fluid volumes (150–160 mL/kg/day), the infant receives the exact enteral accretion recommended by ESPGHAN 2022 across all macronutrients, electrolytes, and bone minerals.

Delivered Energy

119.6 kcal/kg/d

at 150 mL/kg/day

ESPGHAN Target: 115–140 kcal/kg/d. Satisfies resting metabolic energy expenditure.

True Protein (60:40)

3.63 g/kg/d

Whey:Casein Ratio

ESPGHAN Target: 3.5–4.0 g/kg/d. α-lactalbumin promotes lean muscle accretion.

P:E Ratio

3.04 g / 100 kcal

Optimal Anabolic Range

ESPGHAN Target: 2.8–3.6 g/100 kcal. Promotes nitrogen retention without acidosis.

Total Lipids

7.32 g/kg/d

>60% Milk Fat + LCPUFA

ESPGHAN Target: 4.8–8.1 g/kg/d. Easily digested milk fat + DHA/ARA (1:1).

PEDIAMIL® LBW SPECIFICATION VS. ESPGHAN 2022 PRETERM RECOMMENDATIONS

NUTRIENT / ELECTROLYTE	PEDIAMIL® LBW (/100G POWDER)	PEDIAMIL® LBW (/100 ML FEED)	DELIVERED DOSE (150 ML/KG/D)	DELIVERED DOSE (160 ML/KG/D)	ESPGHAN 2022 PRETERM TARGET	COVERAGE
Energy	531 kcal	79.7 kcal	119.6 kcal/kg/d	127.5 kcal/kg/d	115–140 kcal/kg/d (Cond. 140–160)	TARGET MET ✓
True Protein (60:40)	16.1 g (9.66g Whey)	2.42 g (1.45g Whey)	3.63 g/kg/d	3.87 g/kg/d	3.5–4.0 g/kg/d (Cond. up to 4.5)	TARGET MET ✓
Protein-to-Energy (P:E)	3.04 g/100 kcal	3.04 g/100 kcal	3.04 g/100 kcal	3.04 g/100 kcal	2.8–3.6 g / 100 kcal	OPTIMAL RANGE ✓
Total Lipids (Fat)	32.5 g (Milk Fat)	4.88 g	7.32 g/kg/d	7.81 g/kg/d	4.8–8.1 g/kg/d	TARGET MET ✓
DHA & ARA (1:1 Ratio)	90 mg / 90 mg	13.5 mg / 13.5 mg	20.3 mg/kg/d each	21.6 mg/kg/d each	12–30 mg/100 kcal (1:1 Balanced)	BALANCED 1:1 ✓
Calcium (Ca^{2+})	620 mg	93.0 mg (2.32 mmol)	139.5 mg/kg/d	148.8 mg/kg/d	120–140 mg/kg/d (3.0–3.5 mmol/kg/d)	TARGET MET ✓
Phosphorus (P)	355 mg	53.3 mg (1.72 mmol)	80.0 mg/kg/d	85.3 mg/kg/d	60–90 mg/kg/d (Ca:P 1.6–2.0:1)	CA:P 1.75:1 ✓
Elemental Iron (Fe)	10.5 mg	1.58 mg	2.37 mg/kg/d	2.53 mg/kg/d	2.0–3.0 mg/kg/d (Prophylactic Fe)	ENTERAL FE MET ✓
Vitamin D3	11.1 µg (444 IU)	1.66 µg (66.4 IU)	99.6 IU/kg/d	106.2 IU/kg/d	400–1,000 IU/day absolute intake	ADD DROPS IF <400 IU
Sodium (Na^{+})	320 mg	48.0 mg (2.09 mmol)	3.13 mmol/kg/d	3.34 mmol/kg/d	3.0–5.0 mmol/kg/d (ELBW up to 8.0)	ELECTROLYTE MET ✓
Potassium (K^{+})	575 mg	86.3 mg (2.21 mmol)	3.31 mmol/kg/d	3.53 mmol/kg/d	2.0–3.5 mmol/kg/d (78–137 mg/kg/d)	ELECTROLYTE MET ✓
Chloride (Cl^{-})	500 mg	75.0 mg (2.12 mmol)	3.17 mmol/kg/d	3.39 mmol/kg/d	3.0–5.0 mmol/kg/d (106–177 mg/kg/d)	ELECTROLYTE MET ✓
Magnesium (Mg^{2+})	50 mg	7.5 mg	11.3 mg/kg/d	12.0 mg/kg/d	8.0–15.0 mg/kg/d	TARGET MET ✓
Zinc (Zn^{2+})	6.0 mg	0.90 mg	1.35 mg/kg/d	1.44 mg/kg/d	1.4–2.5 mg/kg/d	TARGET MET ✓

Clinical Takeaway: Standalone Enteral Adequacy Without Extra Additives

Pediamil® LBW satisfies ESPGHAN 2022 recommendations at standard physiological volumes (150–160 mL/kg/d). **Doctors do not need to add routine standalone protein fortifiers or electrolyte modulars**, minimizing contamination and osmolar risks in the NICU.

THE ONE-CLICK HOSPITAL FEEDING ORDER SHEET (PDF DOWNLOAD)

Designed specifically for attending neonatologists and pediatric feeding teams, the application embeds a downloadable, printable 7-section clinical worksheet for hospital records, nursing handoff, and pharmacy verification.

OFFICIAL 7-SECTION HOSPITAL CLINICAL ORDER FORM CONTAINS:

PHYSICIAN SIGN-OFF READY

- **Section 1 — Institutional Header:** Hospital emblem, Liptis Nutrition logo, and medical protocol title.
- **Section 2 — Patient Demographics:** Weight (g/kg), Gestational Age, Chronological Age, and Corrected Age.
- **Section 3 — Physician Order:** Prescribed fluid allowance (mL/kg/d) and calculated 24h total feed volume.
- **Section 4 — Preparation Recipe:** Total powder mass (g), exact scoops, sterile water volume, and dilution.
- **Section 5 — Bedside Feeding Timetable:** Exact mL per feed for q3h (8 feeds), q2h (12 feeds), or continuous pump.
- **Section 6 — ESPGHAN 2022 Accretion Table:** Delivered energy, protein, lipids, Ca, P, Fe, and vitamins.
- **Section 7 — Medical Attestation Block:** Attending physician signature line, stamp box, date, time, and medical license number.

AUTHORITATIVE PEER-REVIEWED EVIDENCE REGISTRY

CLINICAL GUIDELINE / STANDARD	PRIMARY CITATION & PUBLICATION	KEY EVIDENCE ROLE IN NEOPED™	VERIFIED INDEX
ESPGHAN 2022 Preterm Enteral Position Paper	Embleton ND, Moltu SJ, Lapillonne A, et al. *J Pediatr Gastroenterol Nutr*. 2023;76(2):248–268.	Defines all enteral energy (115–140 kcal/kg/d), protein (3.5–4.5 g/kg/d), lipid, mineral, and electrolyte bounds.	PubMed PMID: 36705703
Fenton 2013 Preterm Growth Curves	Fenton TR, Kim JH. *BMC Pediatrics*. 2013;13:59.	Intrauterine preterm growth reference curves from 22 to 50 weeks PMA with exact LMS parameters.	PubMed PMID: 23601190
WHO 2006 Child Growth Standards	WHO Multicentre Growth Reference Study Group. *Acta Paediatr Suppl*. 2006;450:76–85.	International gold-standard growth trajectories for term-equivalent infants from 0 to 24 months Corrected Age.	PubMed PMID: 16817681
Cole & Green (1992) LMS Method	Cole TJ, Green PJ. *Stat Med*. 1992;11(10):1305–1319.	Mathematical Box-Cox power transformation for precise age-adjusted Z-score and percentile derivation.	PubMed PMID: 1518992
Patel et al. (2005) Exponential Growth Velocity	Patel AL, Engstrom JL, Meier PP, et al. *J Pediatr*. 2005;147(4):556–561.	Mathematical 2-point exponential model [1000 × $\ln(W_2/W_1)]/(D_2-D_1)$ for accurate daily weight gain velocity in g/kg/day.	PubMed PMID: 16253578
PediTools Independent Benchmarks	Chou JH, Roumiantsev S, Singh R. *J Med Internet Res*. 2020;22(1):e16204.	Independent clinical verification benchmark used to validate NeoPed™ LMS percentile calculations.	PubMed PMID: 32012066
WHO/FAO PIF Reconstitution Guidelines	World Health Organization & FAO (Geneva, 2007) ISBN 978-92-4-159541-4.	Safe reconstitution standards (>70°C sterile water) for powdered infant formula in healthcare facilities.	WHO Item 9789241595414
IUPAC Standard Atomic Weights	Prohaska T, et al. (CIAAW) *Pure Appl Chem*. 2022;94(5):573–600.	Standard atomic weights for stoichiometric molar conversions (Na 22.99, K 39.10, Cl 35.45, Ca 40.08, P 30.97).	DOI: 10.1515/pac-2019-0603

Institutional Medical Governance & Clinical Decision Support Notice:
Liptis Nutrition NeoPed™ is an institutional Clinical Decision Support (CDS) reference calculator intended solely to assist licensed neonatologists, pediatricians, and clinical nutritionists in evaluating enteral feeding regimens. It does not replace individualized clinical judgment, bedside physiological monitoring, or hospital laboratory evaluations. All calculations are executed 100% locally within the user's browser runtime with zero network egress of patient identifiable data.

PREPARED FOR KEY OPINION LEADERS & NICU FACULTY BY:
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CLINICAL VERIFICATION ATTESTATION:
243 / 243 Acceptance Tests Passed (100.0%)
Fully Concordant with ESPGHAN 2022, Fenton 2013 & WHO 2006